$$\langle \sigma_x \rangle = \langle sing | \sigma_x | sing \rangle$$

$$= \quad \langle sing | \sigma_x \frac{1}{\sqrt{2}} \big(|ud\rangle - |du\rangle \big)$$

or

$$\langle \sigma_x \rangle = \frac{1}{2} \Big(\langle ud| - \langle du| \Big) \Big(|dd\rangle - |uu\rangle \Big).$$

Again, this equation gives us zero. Finally, let's look at $\langle \sigma_y \rangle$:

$$\langle \sigma_y \rangle = \langle sing | \sigma_y | sing \rangle$$

$$= \quad \frac{1}{2} \Big(\langle ud| - \langle du| \Big) \Big(i|dd\rangle + i|uu\rangle \Big).$$

As you may have guessed, we are left with zero once more. Thus, we have shown that for the state $|sing\rangle$,

$$\langle \sigma_z \rangle = \langle \sigma_x \rangle = \langle \sigma_y \rangle = 0,$$

and indeed all expectation values of σ are zero. Needless to say, the same is true for the expectation values of τ. Clearly, $|sing\rangle$ is very different from a product state. What does all this say about the measurements we can make?

If the expectation value of a component of σ is zero, it means that the experimental outcome is equally likely to be +1 or −1. In other words, the outcome is completely uncertain. Even though we know the exact state-vector, $|sing\rangle$, we know nothing at all about the outcome of any measurement of any component of either spin.